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Introducing the Chain Conditions

Here we define in their most general form the concepts studied in this monograph, and we record numerous basic and elementary relationships and facts concerning them. The properties are, in increasing order of generality: calibre, compact-calibre, and pseudo-compactness.

On the basis of this work it becomes natural to associate with each space X , for $\lambda > 0$, a cardinal number we call the λ -Souslin number of X (denoted $S_\lambda(X)$). We show (Theorem 2.14, a generalization of a classical result of Erdős & Tarski) that for infinite Hausdorff spaces X and regular $\lambda \geq \omega$, the number $S_\lambda(X)$ is a regular cardinal.

For the most part we deal with arbitrary topological spaces (not subjected to specialized separation conditions), but it is convenient to investigate certain questions about chain conditions in the context of compact Hausdorff spaces (see Chapter 5). The device used for effecting this reduction (with no loss of generality) is the Gleason space of an (arbitrary) space; our treatment of the Gleason space and its definition follows recent work of Galvin.

Chain conditions

Definition. Let X be a space and let α, β and γ be cardinals with $\alpha \geq \beta \geq \gamma$. Then

(a) X has *calibre* (α, β, γ) if for every set $\{U_\xi : \xi < \alpha\}$ of non-empty, open subsets of X there is $B \subset \alpha$ with $|B| = \beta$ such that if $C \subset B$ and $|C| = \gamma$ then $\bigcap_{\xi \in C} U_\xi \neq \emptyset$; and

(b) X has *compact-calibre* (α, β, γ) if for every set $\{U_\xi : \xi < \alpha\}$ of non-empty, open subsets of X there is $B \subset \alpha$ with $|B| = \beta$ such that if $C \subset B$ and $|C| = \gamma$ there is a compact subset K of X such that $U_\xi \cap K \neq \emptyset$ whenever $\xi \in C$.

We consider in practice almost exclusively the case $\alpha \geq \omega$, with $\beta = \alpha$ or $\beta = \gamma$. If X has calibre (α, β, β) we say that X has calibre (α, β) ; and if X has calibre (α, α) we say that X has calibre α (or, α is a calibre for X). The terms *compact-calibre* (α, β) and *compact-calibre* α are defined analogously.

Definition. Let X be a space and let α and β be cardinals with $\alpha \geq \beta$. Then X is *pseudo- (α, β) -compact* if for every set $\{U_\xi : \xi < \alpha\}$ of non-empty, open subsets of X there is $x \in X$ such that

$$|\{\xi < \alpha : V \cap U_\xi \neq \emptyset\}| \geq \beta$$

for every neighborhood V of x .

An indexed family $\{A_i : i \in I\}$ of subsets of X is said to be *locally finite* (in X) if for every $x \in X$ there is a neighborhood V of x such that

$$|\{i \in I : V \cap A_i \neq \emptyset\}| < \omega.$$

Hence X is pseudo- (α, ω) -compact if and only if no locally finite family of non-empty open subsets of X is indexed by α . A pseudo- (α, ω) -compact space is said to be *pseudo- α -compact*, and a pseudo- ω -compact space is said to be *pseudocompact*.

For later use we record in 2.1–2.5 several elementary facts of marginal intrinsic interest about the properties defined above.

2.1 Theorem. Let $\alpha \geq \beta \geq \gamma$, $\alpha \geq \omega$, and X a space.

(a) If $\alpha' \geq \beta' \geq \gamma'$ with $\alpha' \geq \alpha$, $\beta' \leq \beta$ and $\gamma' \leq \gamma$, and if X has calibre (α, β, γ) , then X has calibre $(\alpha', \beta', \gamma')$;

(b) if $\alpha' \geq \beta' \geq \gamma'$ with $\alpha' \geq \alpha$, $\beta' \leq \beta$ and $\gamma' \leq \gamma$, and if X has compact-calibre (α, β, γ) , then X has compact-calibre $(\alpha', \beta', \gamma')$;

(c) if $\alpha' \geq \alpha \geq \beta \geq \beta'$ and X is pseudo- (α, β) -compact, then X is pseudo- (α', β') -compact;

(d) if X has calibre (α, β, γ) , then X has compact-calibre (α, β, γ) ;

(e) if X has calibre (α, β, γ) , then X is pseudo- (α, γ) -compact;

(f) if $\gamma \geq \omega$ and X has compact-calibre (α, β, γ) , then X is pseudo- (α, γ) -compact;

(g) X has compact-calibre (α, α, n) for all $n < \omega$;

(h) if X has calibre $(\alpha, 2)$, then X has calibre (α, n) for all $n < \omega$;

(i) if X is a regular, Hausdorff space and X is pseudo- $(\alpha, 2)$ -compact, then X is pseudo- α -compact.

Proof. (a), (b), and (c) are immediate from the definitions.

(d) follows from the fact that if $x \in \bigcap_{\xi \in C} U_\xi$, then $\{x\}$ is a compact set K such that $U_\xi \cap K \neq \emptyset$ whenever $\xi \in C$.

(e) Let $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(X)$. Since X has calibre (α, γ, γ) there are $C \in [\alpha]^\gamma$ and $x \in X$ such that $x \in \bigcap_{\xi \in C} U_\xi$. It is clear that

$$|\{\xi < \alpha : V \cap U_\xi \neq \emptyset\}| \geq |\{\xi \in C : V \cap U_\xi \neq \emptyset\}| = \gamma$$

for every neighborhood V of x .

(f) If the statement fails there are $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(X)$ and $\{V_x : x \in X\} \subset \mathcal{T}^*(X)$ such that

(i) $x \in V_x$ for $x \in X$, and

(ii) if $A(x) = \{\xi < \alpha : V_x \cap U_\xi \neq \emptyset\}$, then $|A(x)| < \gamma$ for $x \in X$. Since X has compact-calibre (α, γ, γ) , there are $C \in [\alpha]^\gamma$ and compact $K \subset X$ such that $U_\xi \cap K \neq \emptyset$ for all $\xi \in C$. There is $F \in \mathcal{P}_\omega(K)$ such that $K \subset \bigcup_{x \in F} V_x$, so that

$$\gamma = |C| \leq \sum_{x \in F} |C \cap A(x)| \leq \sum_{x \in F} |A(x)| < \gamma,$$

a contradiction.

(g) Let $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(X)$ and $A \in [\alpha]^n$. For $\xi \in A$ let $x_\xi \in U_\xi$ and set $K = \{x_\xi : \xi \in A\}$. Then K is a compact subspace of X such that $U_\xi \cap K \neq \emptyset$ whenever $\xi \in A$.

(h) It follows from part (a) that it is enough to show that if $k < \omega$ and X has calibre (α, k) then X has calibre (α, k^2) . Let $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(X)$ and choose $\{A_\eta : \eta < \alpha\} \subset [\alpha]^k$ so that

$$\alpha = \bigcup_{\eta < \alpha} A_\eta, \text{ and}$$

$$A_\eta \cap A_{\eta'} = \emptyset \text{ for } \eta' < \eta < \alpha.$$

Since X has calibre (α, k) , for $\eta < \alpha$ there is $B_\eta \in [A_\eta]^k$ such that $\bigcap \{U_\xi : \xi \in B_\eta\} \neq \emptyset$; we set $V_\eta = \bigcap \{U_\xi : \xi \in B_\eta\}$ for $\eta < \alpha$. Again since X has calibre (α, k) , there is $C \in [\alpha]^k$ such that $\bigcap_{\eta \in C} V_\eta \neq \emptyset$. We set $D = \bigcup_{\eta \in C} B_\eta$. Then $D \in [\alpha]^{k^2}$, and

$$\bigcap_{\xi \in D} U_\xi = \bigcap_{\eta \in C} V_\eta \neq \emptyset,$$

as required.

(i) If the statement fails there is locally finite $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(X)$. We choose $x_\xi \in U_\xi$ and we note that since

$$|\{\xi < \alpha : x_\xi \in U_\xi\}| < \omega \text{ for } \xi < \alpha$$

there is $A \in [\alpha]^\omega$ such that

$$x_\xi \neq x_\zeta \text{ for } \xi, \zeta \in A, \xi \neq \zeta.$$

For $\xi \in A$ there is a neighborhood V_ξ of x_ξ such that

$$|\{\zeta \in A : V_\xi \cap U_\zeta \neq \emptyset\}| < \omega;$$

we assume without loss of generality, shrinking V_ξ if necessary, that

$$\text{cl } V_\xi \subset U_\xi \text{ and } x_\xi \notin \text{cl } V_\xi \text{ for } \xi, \zeta \in A, \xi \neq \zeta.$$

We set

$$W_\xi = V_\xi \setminus \cup \{\text{cl } V_\zeta : \zeta \in A, \zeta \neq \xi\} \text{ for } \xi \in A.$$

It is easy to verify, using the fact that $\{V_\zeta : \zeta \in A\}$ is a locally finite family in X , that $\cup \{\text{cl } V_\zeta : \zeta \in A, \zeta \neq \xi\}$ is closed in X ; hence we have $W_\xi \in \mathcal{T}^*(X)$ for $\xi \in A$. For $\xi \in A$ we choose $X_\xi \in \mathcal{T}^*(X)$ such that

$$x_\xi \in X_\xi \subset \text{cl } X_\xi \subset W_\xi,$$

and we claim, finally, that for $x \in X$ there is a neighborhood U of x such that $|\{\xi \in A : U \cap X_\xi \neq \emptyset\}| \leq 1$. Indeed if there is $\xi \in A$ such that $x \in \text{cl } X_\xi$, we set $U = W_\xi$, and if $x \notin \cup_{\xi \in A} (\text{cl } X_\xi)$ we set

$$U = X \setminus \cup_{\xi \in A} (\text{cl } X_\xi).$$

The proof is complete.

Theorem 2.1 essentially exhausts the implications which hold among the various topological properties discussed in this chapter (but see Theorem 5.23 below, which improves Theorem 2.1(h)).

In Chapter 3 we use the Erdős–Rado theorems to investigate products of (infinitely many) spaces with the properties defined above. But the following simple result is available immediately from the definitions.

2.2 Theorem. Let $\alpha \geq \beta \geq \gamma \geq \delta \geq 2$ with $\beta \geq \omega$ and let X and Y be spaces.

(a) If X has calibre (α, β, δ) and Y has calibre (β, γ, δ) , then $X \times Y$ has calibre (α, γ, δ) .

(b) If X has compact-calibre (α, β, δ) and Y has compact-calibre (β, γ, δ) , then $X \times Y$ has compact-calibre (α, γ, δ) .

Proof. Let $\{U_\xi \times V_\xi : \xi < \alpha\} \subset \mathcal{T}^*(X) \times \mathcal{T}^*(Y)$. In (a) there is $B \in [\alpha]^\beta$ such that if $D \in [B]^\delta$ then $\cap_{\xi \in D} U_\xi \neq \emptyset$, and in (b) there is

$B \in [\alpha]^\beta$ such that for $D \in [B]^\delta$ there is compact $K \subset X$ such that $U_\xi \cap K \neq \emptyset$ whenever $\xi \in D$. And in (a) there is $C \in [B]^\gamma$ such that if $D \in [C]^\delta$ then $\bigcap_{\xi \in D} V_\xi \neq \emptyset$, and in (b) there is $C \in [B]^\gamma$ such that for $D \in [C]^\delta$ there is compact $L \subset Y$ such that $V_\xi \cap L \neq \emptyset$ whenever $\xi \in D$. It is then clear in (a) that if $D \in [C]^\delta$ then $\bigcap_{\xi \in D} (U_\xi \times V_\xi) \neq \emptyset$, and in (b) that for $D \in [C]^\delta$ there is compact $K \times L \subset X \times Y$ such that $(U_\xi \times V_\xi) \cap (K \times L) \neq \emptyset$ whenever $\xi \in D$.

It follows in particular (replacing α, β, γ and δ by α, α, α and β , respectively) that if $\alpha \geq \beta \geq 2$ with $\alpha \geq \omega$ then the product of a finite number of spaces with calibre (α, α, β) has calibre (α, α, β) and the product of a finite number of spaces with compact-calibre (α, α, β) has compact-calibre (α, α, β) .

We note in Corollary 9.8 that for $\alpha \geq \beta \geq \omega$ the product of a finite number of pseudo- (α, β) -compact spaces, each completely regular and Hausdorff, need not be pseudo- (α, β) -compact.

The pseudoweight $\pi w(X)$ of a space X is defined in the Appendix.

2.3 Theorem. Let $\alpha \geq \omega$, and let X be a space.

- (a) If X has calibre α then X has calibre $\text{cf}(\alpha)$;
- (a') if X has calibre $\text{cf}(\alpha)$ and $\pi w(X) < \alpha$, then X has calibre α ;
- (b) if X has compact-calibre α then X has compact-calibre $\text{cf}(\alpha)$;
- (b') if X has compact-calibre $\text{cf}(\alpha)$ and $\pi w(X) < \alpha$, then X has compact-calibre α ;
- (c) if X is pseudo- (α, α) -compact then X is pseudo- $(\text{cf}(\alpha), \text{cf}(\alpha))$ -compact; and
- (c') if X is pseudo- $(\text{cf}(\alpha), \text{cf}(\alpha))$ -compact and $\pi w(X) < \alpha$, then X is pseudo- (α, α) -compact.

Proof. We assume that α is singular and we choose a family $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ such that

$$0 = \alpha_0 < \alpha_{\sigma'} < \alpha_\sigma < \alpha \text{ for } 0 < \sigma' < \sigma < \text{cf}(\alpha), \text{ and} \\ \alpha = \sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma.$$

We first prove (a), (b) and (c). Let $\{U_\sigma : \sigma < \text{cf}(\alpha)\} \subset \mathcal{T}^*(X)$, and set

$$V_\eta = U_\sigma \text{ for } \alpha_\sigma \leq \eta < \alpha_{\sigma+1}.$$

Then in (a) there is $A \in [\alpha]^\alpha$ such that $\bigcap_{\eta \in A} V_\eta \neq \emptyset$; in (b) there

are $A \in [\alpha]^\alpha$ and compact $K \subset X$ such that $V_\eta \cap K \neq \emptyset$ whenever $\eta \in A$; and in (c) there is $x \in X$ such that $|\{\eta < \alpha : V \cap V_\eta \neq \emptyset\}| = \alpha$ for every neighborhood V of x .

Now in (a) and (b) set

$$A(\sigma) = A \cap [\alpha_\sigma, \alpha_{\sigma+1}) \text{ for } \sigma < \text{cf}(\alpha),$$

and in (c) let V be a neighborhood of x and set

$$A(\sigma) = \{\eta \in [\alpha_\sigma, \alpha_{\sigma+1}) : V \cap V_\eta \neq \emptyset\} \text{ for } \sigma < \text{cf}(\alpha);$$

in each case set $B = \{\sigma < \text{cf}(\alpha) : A(\sigma) \neq \emptyset\}$, and for $\sigma \in B$ let $\eta(\sigma) \in A(\sigma)$. Then $B \in [\text{cf}(\alpha)]^{\text{cf}(\alpha)}$ (since $A = \bigcup_{\sigma \in B} A(\sigma)$ and $|A(\sigma)| < \alpha$ for $\sigma \in B$) and we have:

$$\begin{aligned} \bigcap_{\sigma \in B} U_\sigma &= \bigcap_{\sigma \in B} V_{\eta(\sigma)} \neq \emptyset \text{ in (a),} \\ U_\sigma \cap K &= V_{\eta(\sigma)} \cap K \neq \emptyset \text{ for } \sigma \in B \text{ in (b), and} \\ V \cap U_\sigma &= V \cap N_{\eta(\sigma)} \neq \emptyset \text{ for } \sigma \in B \text{ in (c).} \end{aligned}$$

as required.

We prove (a'), (b') and (c'). Let $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(X)$, let $\{B_\eta : \eta < \beta\}$ be a pseudobase for X with $\beta < \alpha$, and set

$$A(\eta) = \{\xi < \alpha : B_\eta \subset U_\xi\} \text{ for } \eta < \beta.$$

Since $\alpha = \bigcup_{\eta < \beta} A(\eta)$ and $\beta < \alpha$, we have

$$\sup \{|A(\eta)| : \eta < \beta\} = \alpha;$$

for $\sigma < \text{cf}(\alpha)$ we choose $\eta(\sigma) < \beta$ such that $|A(\eta(\sigma))| \geq \alpha_\sigma$. Then in (a') there is $C \in [\text{cf}(\alpha)]^{\text{cf}(\alpha)}$ such that $\bigcap_{\sigma \in C} B_{\eta(\sigma)} \neq \emptyset$; in (b') there are $C \in [\text{cf}(\alpha)]^{\text{cf}(\alpha)}$ and compact $K \subset X$ such that $B_{\eta(\sigma)} \cap K \neq \emptyset$ whenever $\sigma \in C$; and in (c') there is $x \in X$ such that $|\{\sigma < \text{cf}(\alpha) : V \cap B_{\eta(\sigma)} \neq \emptyset\}| = \text{cf}(\alpha)$ for every neighborhood V of x . In (c') let V be a neighborhood of x and set

$$C = \{\sigma < \text{cf}(\alpha) : V \cap B_{\eta(\sigma)} \neq \emptyset\},$$

and in each case set $A = \bigcup_{\sigma \in C} A(\eta(\sigma))$. Then $A \in [\alpha]^\alpha$ and we have:

$$\begin{aligned} \bigcap_{\xi \in A} U_\xi &\supset \bigcap_{\sigma \in C} B_{\eta(\sigma)} \neq \emptyset \text{ in (a'),} \\ U_\xi \cap K &\supset B_{\eta(\sigma)} \cap K \neq \emptyset \text{ for } \xi \in A(\eta(\sigma)) \text{ in (b'), and} \\ V \cap V_\xi &\supset V \cap B_{\eta(\sigma)} \neq \emptyset \text{ for } \xi \in A(\eta(\sigma)) \text{ in (c'),} \end{aligned}$$

as required.

The proof is complete.

We show in Theorems 4.3, 6.2 and 7.1 that for α singular the assumption $\pi w(X) < \alpha$ in (a'), (b') and (c') cannot be omitted.

2.4 Theorem. Let $\alpha \geq \beta \geq \omega$ and let γ be a cardinal number.

(a) The following are equivalent: the space γ has calibre (α, β) ; the space γ has compact-calibre (α, β) ; the space γ is pseudo- (α, β) -compact.

(b) The space γ has calibre α if and only if $\gamma < \text{cf}(\alpha)$.

(c) The space γ has calibre (α, β) if and only if $(\alpha = \beta \text{ and } \gamma < \text{cf}(\alpha))$ or $(\beta < \alpha \text{ and } \gamma < \alpha)$.

Proof. (a) For $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(\gamma)$ and $\eta \in \gamma$ we have ($\{\eta\}$ is a neighborhood of η and) $\eta \in U_\xi$ if and only if $\{\eta\} \cap U_\xi \neq \emptyset$. Thus the discrete space γ , if pseudo- (α, β) -compact, has calibre (α, β) . The remaining required implications are given by Theorem 2.1 ((d) and (f)).

(b) $(\Rightarrow)$ There is $\{A_\sigma : \sigma < \text{cf}(\alpha)\}$ such that

$$\begin{aligned} \bigcup_{\sigma < \text{cf}(\alpha)} A_\sigma &= \alpha, \\ A_{\sigma'} \cap A_\sigma &= \emptyset \quad \text{for } \sigma' < \sigma < \text{cf}(\alpha), \text{ and} \\ |A_\sigma| &< \alpha \quad \text{for } \sigma < \text{cf}(\alpha). \end{aligned}$$

If $\text{cf}(\alpha) \leq \gamma$, then for $\xi \in A_\sigma$ we set

$$U_\xi = \{\sigma\} \subset \text{cf}(\alpha) \subset \gamma.$$

Then $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(\gamma)$ and for $\eta \in \gamma$ the set $\{\eta\}$ is a neighborhood of η such that

$$|\{\xi < \alpha : \{\eta\} \cap U_\xi \neq \emptyset\}| < \alpha,$$

a contradiction.

$(\Leftarrow)$ Clearly for $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(\gamma)$ there is $\eta \in \gamma$ such that $|\{\xi < \alpha : \eta \in U_\xi\}| = \alpha$.

(c) $(\Rightarrow)$ Suppose that γ has calibre (α, β) . If $\alpha = \beta$ then $\gamma < \text{cf}(\alpha)$ by (b). If $\beta < \alpha$ and $\gamma \geq \alpha$, then $\{\{\xi\} : \xi < \alpha\}$ is a family of non-empty open subsets of α (hence of γ) such that

$$\{\xi'\} \cap \{\xi\} = \emptyset \text{ for } \xi' < \xi < \alpha,$$

so that γ does not have calibre (α, β) .

$(\Leftarrow)$ If $\beta = \alpha$ and $\gamma < \text{cf}(\alpha)$, then γ has calibre (α, β) by (b). If $\beta < \alpha$

and $\gamma < \alpha$ we set $\delta = \max \{\beta, \gamma\}$; since $\gamma < \delta^+ = \text{cf}(\delta^+)$ the space γ has calibre δ^+ by (b), and since $\beta < \delta^+ \leq \alpha$ the space γ has calibre (α, β) by Theorem 2.1(a).

We note finally that the properties considered above are preserved under certain standard topological operations. We consider in particular the following four (possible) relationships between spaces X and Y :

- (i) There is a continuous function from Y onto X ; or
- (ii) Y is a dense subspace of X ; or
- (iii) X is an open subspace of Y ; or
- (iv) X is the closure (in Y) of an open subspace of Y .

2.5 Theorem. Let $\alpha \geq \beta \geq \gamma \geq 2$ and let X and Y be spaces.

(a) If X and Y are related as in (i), (ii), (iii) or (iv) above and if Y has calibre (α, β, γ) , then X has calibre (α, β, γ) ;

(b) if X and Y are related as in (i), (ii) or (iv) above and if Y has compact-calibre (α, β, γ) or is pseudo- (α, β) -compact, then X has the same property.

Proof. We note that statement (a)(iv) follows from (a)(ii) and (a)(iii).

We prove that if Y is pseudo- (α, β) -compact and $X = \text{cl}_Y U$ with $U \in \mathcal{T}(Y)$, then X is pseudo- (α, β) -compact. If $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(X)$, then since $\{U \cap U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(Y)$ there is $x \in Y$ such that

$$|\{\xi < \alpha : V \cap (U \cap U_\xi) \neq \emptyset\}| \geq \beta$$

for every neighborhood V (in Y) of x . If $x \in Y \setminus X$, then $Y \setminus X$ is a neighborhood in Y of x such that $(Y \setminus X) \cap (U \cap U_\xi) = \emptyset$ for all $\xi < \alpha$; hence $x \in X$, as required.

We omit the remaining routine verifications.

2.6 Remarks. (a) It follows from Theorem 2.5((i) and (ii)) that if $\alpha \geq \beta \geq \gamma \geq 2$ and X is a space, and if the discrete space $d(X)$ has calibre (α, β, γ) or compact-calibre (α, β, γ) or is pseudo- (α, β) -compact, then X has the same property.

(b) There is a converse to (a portion of) Theorem 2.5(a)(ii): If $\alpha \geq \beta \geq 2$ and Y is dense in X and X has calibre $(\alpha, \beta, 2)$, then Y has calibre $(\alpha, \beta, 2)$. If $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(Y)$, there is $\{\tilde{U}_\xi : \xi < \alpha\}$ such

that $\tilde{U}_\xi \cap Y = U_\xi$ for $\xi < \alpha$, and since X has calibre $(\alpha, \beta, 2)$ there is $B \in [\alpha]^\beta$ such that $\tilde{U}_\xi \cap \tilde{U}_{\xi'} \neq \emptyset$ for $\xi, \xi' \in B$; since Y is dense in X it follows that $U_\xi \cap U_{\xi'} \neq \emptyset$ for $\xi, \xi' \in B$, as required.

(c) If $\alpha \geq \beta \geq \gamma$ and $\alpha \geq \omega$, then every compact space has compact-calibre (α, β, γ) and is pseudo- (α, β) -compact. Hence an open subspace of a space with compact-calibre (α, β, γ) need not have compact-calibre (α, β, γ) , and an open subspace of a pseudo- (α, β) -compact space need not be pseudo- (α, β) -compact.

We note next that for every non-empty space X some (infinite, regular) cardinal is a calibre of X .

2.7 Theorem. Let α be an infinite cardinal and X a non-empty space. If $d(X) < \text{cf}(\alpha)$, then X has calibre α .

Proof. Let $\{x_\eta : \eta < \beta\}$ be a dense subset of X with $\beta < \text{cf}(\alpha)$, let $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(X)$, and set

$$A(\eta) = \{\xi < \alpha : x_\eta \in U_\xi\} \text{ for } \eta < \beta.$$

Since $\bigcup_{\eta < \beta} A(\eta) = \alpha$ and $\beta < \text{cf}(\alpha)$, there is $\bar{\eta} < \beta$ such that $A(\bar{\eta}) \in [\alpha]^\alpha$; we have

$$x_{\bar{\eta}} \in \bigcap_{\xi \in A(\bar{\eta})} U_\xi,$$

as required.

The λ -Souslin numbers

Definition. Let X be a space. A *cellular family* in X is an indexed set $\{U_i : i \in I\}$ of non-empty open subsets of X such that

$$U_i \cap U_{i'} = \emptyset \text{ for } i, i' \in I, i \neq i'.$$

It follows from Theorem 2.7 that for X a non-empty space and λ a cardinal, $\lambda > 0$, there is a cardinal α such that X has calibre (α, λ) . (One may take $\alpha = (\max\{d(X), \lambda, \omega\})^+$.) This validates the following definition.

Definition. For $\lambda > 0$ and X a non-empty space, the λ -Souslin number of X , denoted $S_\lambda(X)$, is the cardinal

$$S_\lambda(X) = \min\{\alpha : X \text{ has calibre } (\alpha, \lambda)\}.$$

The 2-Souslin number of X is called, simply, the *Souslin number* of X and is (usually) denoted $S(X)$.

We note some simple consequences of this definition.

2.8 Lemma. Let $\lambda > 0$, let α be a cardinal and let X be a non-empty space.

(a) $\lambda \leq S_\lambda(X)$;

(b) $\alpha < S_\lambda(X)$ if and only if there is a set $\{U_\xi : \xi < \alpha\}$ of non-empty, open subsets of X such that if $A \subset \alpha$ with $|A| = \lambda$ then $\bigcap_{\xi \in A} U_\xi = \emptyset$;

(c) if $\lambda = 1$ then $S_\lambda(X) = 1$;

(d) if $\lambda = 2$ then $S_\lambda(X)$ is the least cardinal β such that no cellular family in X is indexed by β ; and,

(e) if $\lambda \geq \kappa$ then $S_\lambda(X) \geq S_\kappa(X)$.

We determine the λ -Souslin numbers of discrete spaces.

2.9 Lemma. Let $\lambda > 0$ and $\beta > 0$ and set

$$\begin{aligned} \alpha &= (\lambda - 1)\beta + 1 && \text{if } \lambda < \omega, \beta < \omega, \\ \alpha &= \beta^+ && \text{if } \beta \geq \omega, \beta \geq \lambda > 1, \\ \alpha &= \lambda^+ && \text{if } \lambda \geq \omega, \text{cf}(\lambda) \leq \beta < \lambda, \\ \alpha &= \lambda && \text{if } \lambda \geq \omega, \beta < \text{cf}(\lambda). \end{aligned}$$

Then $\alpha = S_\lambda(\beta)$.

Proof. ($\geq$) It is clear in each case that if $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(\beta)$ and $x(\xi) \in U_\xi$, then there is $\eta < \beta$ such that

$$|\{\xi < \alpha : x(\xi) = \eta\}| \geq \lambda.$$

($\leq$) For $\lambda \geq \omega, \beta < \text{cf}(\lambda)$ we have $S_\lambda(\beta) \geq \lambda = \alpha$ from Lemma 2.8(a). To establish the required inequality in the other three cases we set

$$\begin{aligned} \gamma &= (\lambda - 1)\beta && \text{if } \lambda < \omega, \beta < \omega \\ \gamma &= \beta && \text{if } \beta \geq \omega, \beta \geq \lambda > 1 \\ \gamma &= \lambda && \text{if } \lambda \geq \omega, \text{cf}(\lambda) \leq \beta < \lambda, \end{aligned}$$

and we note in each case that it is simple to find $\{x(\xi) : \xi < \gamma\} \subset \beta$ so that

$$|\{\xi < \gamma : x(\xi) = \eta\}| < \lambda \text{ for } \eta < \beta.$$

It follows that $S_\lambda(\beta) > \gamma$; hence $S_\lambda(\beta) \geq \alpha$.

The following terminology will be used (only) in the proof of Lemma 2.10 below.

A space X has *large finite cellular families* if for all $n < \omega$ there is a cellular family $\{U_i : i \in I\}$ in X such that $|I| \geq n$.

It is clear that if X has large finite cellular families and $\mathcal{U}$ is a finite family of open subsets of X such that $\bigcup \mathcal{U}$ is dense in X , then there is $U \in \mathcal{U}$ such that U has large finite cellular families.

2.10 Lemma. Let X be a space. If X has large finite cellular families, then there is a cellular family $\{U_n : n < \omega\}$ (indexed by ω) in X .

Proof. For $n < \omega$ let $\mathcal{U}_n = \{U_i(n) : i \in I(n)\}$ be a cellular family in X with $n + 2 \leq |I(n)| < \omega$. We assume without loss of generality, adjoining

$$X \setminus \text{cl}(\bigcup_{i \in I(n)} U_i(n))$$

to $\mathcal{U}_n$ and reindexing, that $\bigcup \mathcal{U}_n$ is dense in X for $n < \omega$. We assume also, recursively replacing $\mathcal{U}_{n+1}$ by

$$\{U_i(n+1) \cap U_j(n) : i \in I(n+1), j \in I(n), U_i(n+1) \cap U_j(n) = \emptyset\}$$

and reindexing, that $\mathcal{U}_{n+1}$ is a *refinement* of $\mathcal{U}_n$ for $n < \omega$ (in the sense that for $i \in I(n+1)$ there is $j \in I(n)$ such that $U_i(n+1) \subset U_j(n)$). Finally we assume, again using finite recursion to replace $\mathcal{U}_n$ by a refinement, that if $i \in I(n)$ and $U_i(n)$ has large finite cellular families, then

$$|\{j \in I(n+1) : U_j(n+1) \subset U_i(n)\}| \geq 2.$$

It is now simple to define by finite recursion sets $\{i(n) : n < \omega\}$ and $\{j(n) : n < \omega\}$ such that

- (i) $i(n), j(n) \in I(n)$ for $n < \omega$;
- (ii) $U_{i(n)}(n)$ has large finite cellular families for $n < \omega$;
- (iii) $i(n) \neq j(n)$ for $n < \omega$; and
- (iv) $U_{i(n)}(n) \cup U_{j(n)}(n) \subset U_{i(n-1)}(n-1)$ for $0 < n < \omega$.

It is then clear that $\{U_{j(n)}(n) : n < \omega\}$ is a cellular family in X .

The proof of Lemma 2.10 is essentially the proof of the classical tree theorem of D. König (1927).

From Lemma 2.10 it follows that if X is a space then $S(X) \neq \omega$.

2.11 Corollary. Let X be an infinite Hausdorff space and $1 < \lambda < \omega$. Then

(a) $S_2(X) \geq \omega^+$ (i.e., X does not have calibre $(\omega, 2, 2)$);

(b) $S_2(X) = S_\lambda(X)$.

Proof. (a) It follows from Lemma 2.10 that there is a cellular family in X indexed by ω . Hence $S_2(X) \geq \omega^+$.

(b) now follows from Theorem 2.1((a) and (h)).

2.12 Corollary. (a) $\{0, 1\}^m$ has calibre α for $0 < m < \omega, \alpha \geq \omega$; and

(b) $\{0, 1\}^\omega$ does not have calibre α if $\text{cf}(\alpha) = \omega$.

Proof. (a) Since $|\{0, 1\}^m| < \omega$ for $0 < m < \omega$, this follows from Theorem 2.6.

(b) From Corollary 2.11(a), Theorem 2.1(a) and Theorem 2.3(a), respectively, it follows that $\{0, 1\}^\omega$ does not have calibre $(\omega, 2, 2)$, hence does not have calibre ω , hence does not have calibre α if $\text{cf}(\alpha) = \omega$.

2.13 Lemma. Let $\lambda \geq 2$ and let X be a non-empty space. If $S_\lambda(X)$ is a limit cardinal, then there is a non-empty, open subset U of X such that $S_\lambda(V) = S_\lambda(X)$ for every non-empty, open subset V of U .

Proof. From Lemma 2.8(a) we have $S_\lambda(X) \geq \lambda$. If $S_\lambda(X) = \lambda$, then for $V \in \mathcal{T}^*(X)$ we have

$$\lambda \leq S_\lambda(V) \leq S_\lambda(X) \leq \lambda,$$

and we complete the proof by setting $U = X$. In what follows we assume $S_\lambda(X) > \lambda$.

If the lemma fails, then for all $U \in \mathcal{T}^*(X)$ there is $V \in \mathcal{T}^*(U)$ such that $S_\lambda(V) < S_\lambda(X)$. We choose a maximal cellular family $\{V_i : i \in I\}$ of X such that $S_\lambda(V_i) < S_\lambda(X)$ for $i \in I$, and we set $\alpha = \sum_{i \in I} S_\lambda(V_i)$. Since $|I| < S_2(X) \leq S_\lambda(X)$, we have

$$\lambda \leq \alpha \leq |I| \cdot S_\lambda(X) = S_\lambda(X);$$

we claim $\alpha < S_\lambda(X)$.

If the claim fails, then α is the limit cardinal $S_\lambda(X)$ and there is a family $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ of cardinal numbers such that

$$\lambda = \alpha_0 \leq \alpha_{\sigma'} < \alpha_\sigma < \alpha \quad \text{for } \sigma' < \sigma < \text{cf}(\alpha), \text{ and}$$

$$\sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma = \alpha.$$

We note that $\alpha = \sup \{S_\lambda(V_i) : i \in I\}$; indeed if there is $\beta < \alpha$ such that $S_\lambda(V_i) \leq \beta$ for all $i \in I$, then from $|I| < S_2(X) \leq S_\lambda(X) = \alpha$ we would have the contradiction

$$\alpha = \sum_{i \in I} S_\lambda(V_i) \leq |I| \cdot \beta < \alpha.$$

Thus for $\sigma < \text{cf}(\alpha)$ there is $i(\sigma) \in I$ such that $S_\lambda(V_{i(\sigma)}) > \alpha_{\sigma+1} > \lambda$. We define $\{U_\xi : \alpha_\sigma \leq \xi < \alpha_{\sigma+1}\} \subset \mathcal{T}^*(V_{i(\sigma)})$ such that $\cap_{\xi \in A} U_\xi = \emptyset$ for all $A \in [\alpha_{\sigma+1} \setminus \alpha_\sigma]^\lambda$ and we note that if $A \in [\alpha \setminus \lambda]^\lambda$ then $\cap_{\xi \in A} U_\xi = \emptyset$; this is clear if there is $\sigma < \text{cf}(\alpha)$ such that $A \in [\alpha_{\sigma+1} \setminus \alpha_\sigma]^\lambda$, while if there are $\zeta, \zeta' \in A$ and $\sigma, \sigma' < \text{cf}(\alpha)$ such that

$$\alpha_{\sigma'} \leq \zeta' < \alpha_{\sigma'+1} \text{ and } \alpha_\sigma \leq \zeta < \alpha_{\sigma+1},$$

then $\cap_{\xi \in A} U_\xi \subset U_{\zeta'} \cap U_\zeta \subset V_{i(\sigma')} \cap V_{i(\sigma)} = \emptyset$.

This completes the proof that $S_\lambda(X) > \alpha$.

Now set $\beta = (\alpha \cdot |I|)^+$. Since $\beta < S_\lambda(X)$, there is $\{W_\xi : \xi < \beta\} \subset \mathcal{T}^*(X)$ such that $\cap_{\xi \in A} W_\xi = \emptyset$ for all $A \in [\beta]^\lambda$. For $i \in I$ we set

$$A_i = \{\xi < \beta : W_\xi \cap V_i \neq \emptyset\}.$$

From the maximality of the cellular family $\{V_i : i \in I\}$ it follows that $\cup_{i \in I} V_i$ is dense in X , so that $\cup_{i \in I} A_i = \beta$. Since $|I| < \beta$ there is $\bar{i} \in I$ such that $|A_{\bar{i}}| = \beta$, and from the relations

$$\begin{aligned} \{W_\xi \cap V_{\bar{i}} : \xi \in A_{\bar{i}}\} &\subset \mathcal{T}^*(V_{\bar{i}}), \\ S_\lambda(V_{\bar{i}}) &\leq \alpha < \beta \end{aligned}$$

there is $A \in [A_{\bar{i}}]^\lambda$ such that $\cap_{\xi \in A} (W_\xi \cap V_{\bar{i}}) \neq \emptyset$. We have

$$\emptyset \neq \bigcap_{\xi \in A} (W_\xi \cap V_{\bar{i}}) \subset \bigcap_{\xi \in A} W_\xi = \emptyset,$$

a contradiction.

The proof is complete.

2.14 Theorem. Let λ be a cardinal such that either $\lambda = 2$ or λ is (infinite and) regular, and let X be an infinite Hausdorff space. Then $S_\lambda(X)$ is a regular cardinal.

Proof. We argue by contradiction. We assume the result fails, we set $\alpha = S_\lambda(X)$, we note that $\lambda < \alpha$ and we (will) define $\{V_\xi : \lambda \leq \xi < \alpha\} \subset \mathcal{T}^*(X)$ such that $\cap_{\xi \in A} V_\xi = \emptyset$ for all $A \in [\alpha \setminus \lambda]^\lambda$.

From Corollary 2.11 we have $S(X) \geq \omega^+$. Since $\alpha = S_\lambda(X) \geq S(X)$ (by Lemma 2.8(e)) the cardinal α is a singular limit cardinal and from

Lemma 2.13 there is $U \in \mathcal{T}^*(X)$ such that $S_\lambda(V) = \alpha$ for all $V \in \mathcal{T}^*(U)$. We choose a family $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ of cardinal numbers such that

$$\lambda = \alpha_0 \leq \alpha_{\sigma'} < \alpha_\sigma < \alpha \text{ for } \sigma' < \sigma < \text{cf}(\alpha), \text{ and} \\ \sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma = \alpha,$$

and we define $\{V_\xi : \alpha_\sigma \leq \xi < \alpha_{\sigma+1}\}$ for $\sigma < \text{cf}(\alpha)$ as follows. If $\text{cf}(\alpha) < \lambda$ we define $\{V_\xi : \alpha_\sigma \leq \xi < \alpha_{\sigma+1}\} \subset \mathcal{T}^*(U)$ so that $\bigcap_{\xi \in A} V_\xi = \emptyset$ for all $A \in [\alpha_{\sigma+1} \setminus \alpha_\sigma]^\lambda$; and if $\lambda \leq \text{cf}(\alpha)$ we define $\{U_\sigma : \sigma < \text{cf}(\alpha)\} \subset \mathcal{T}^*(U)$ so that $\bigcap_{\sigma \in A} U_\sigma = \emptyset$ for all $A \in [\text{cf}(\alpha)]^\lambda$ and (using the relation $S_\lambda(U_\sigma) = \alpha$) we define $\{V_\xi : \alpha_\sigma \leq \xi < \alpha_{\sigma+1}\} \subset \mathcal{T}^*(U_\sigma)$ so that $\bigcap_{\xi \in A} V_\xi = \emptyset$ for all $A \in [\alpha_{\sigma+1} \setminus \alpha_\sigma]^\lambda$.

We show that the family $\{V_\xi : \lambda \leq \xi < \alpha\}$ is as required. Let $A \in [\alpha \setminus \lambda]^\lambda$, set

$$A_\sigma = A \cap [\alpha_\sigma, \alpha_{\sigma+1}) \text{ for } \sigma < \text{cf}(\alpha),$$

and set $B = \{\sigma < \text{cf}(\alpha) : A_\sigma \neq \emptyset\}$. If $|B| < \lambda$, then since $\lambda = 2$ or λ is regular there is $\bar{\sigma} < \text{cf}(\alpha)$ such that $|A_{\bar{\sigma}}| = \lambda$ and we have

$$\bigcap_{\xi \in A} V_\xi \subset \bigcap \{V_\xi : \xi \in A_{\bar{\sigma}}\} = \emptyset;$$

and if $|B| = \lambda$ then ($\lambda \leq \text{cf}(\alpha)$ and U_σ is defined for $\sigma < \text{cf}(\alpha)$ and) we have

$$\bigcap_{\xi \in A} V_\xi \subset \bigcap_{\sigma \in B} U_\sigma = \emptyset.$$

Reduction to compact Hausdorff spaces

For use in describing the calibre properties of compact spaces we introduce the relations $\leq$ and $\equiv$ on the class of spaces and we show that every space X is $\equiv$ -equivalent to its (compact) Gleason space $G(X)$.

Definition. Let X and Y be spaces. Then $X \leq Y$ if there is $\varphi : \mathcal{T}^*(X) \rightarrow \mathcal{T}^*(Y)$ such that

if $n < \omega$ and $\{U_k : k < n\} \subset \mathcal{T}^*(X)$ and $\bigcap_{k < n} U_k = \emptyset$, then $\bigcap_{k < n} \varphi(U_k) = \emptyset$.

With X, Y and φ as above we say that φ makes $X \leq Y$.

It is clear that if X, Y and Z are spaces, then $X \leq X$ (i.e., $\leq$ is a

reflexive relation) and if $X \leq Y$ and $Y \leq Z$ then $X \leq Z$ (i.e., $\leq$ is a transitive relation).

2.15 Lemma. Let X and Y be spaces, $\mathcal{B}$ a pseudobase for X and φ a function from $\mathcal{B}$ into $\mathcal{T}^*(Y)$ such that

if $n < \omega$ and $\{B_k : k < n\} \subset \mathcal{B}$ and $\bigcap_{k < n} B_k = \emptyset$, then $\bigcap_{k < n} \varphi(B_k) = \emptyset$. Then $X \leq Y$.

Proof. For $U \in \mathcal{T}^*(X)$ we choose $B_U \in \mathcal{B}$ with $\emptyset \neq B_U \subset U$, and we define $\psi : \mathcal{T}^*(X) \rightarrow \mathcal{T}^*(Y)$ by the rule $\psi(U) = \varphi(B_U)$. Then ψ makes $X \leq Y$.

2.16 Theorem. Let X and Y be spaces.

- (a) If X is an open subspace of Y , then $X \leq Y$;
- (b) if X is a dense subspace of Y , then $X \leq Y$ and $Y \leq X$;
- (c) if X is a continuous image of Y , then $X \leq Y$; and
- (d) if there are sets $\{X_i : i \in I\}$ and $\{Y_i : i \in I\}$ of spaces with $X_i \leq Y_i$ for $i \in I$ and with $X = \prod_{i \in I} X_i$ and $Y = \prod_{i \in I} Y_i$, then $X \leq Y$.

Proof. (a) Define $\varphi : \mathcal{T}^*(X) \rightarrow \mathcal{T}^*(Y)$ by $\varphi(U) = U$. Then φ makes $X \leq Y$.

(b) For $U \in \mathcal{T}^*(X)$ let $\varphi(U)$ be an element of $\mathcal{T}^*(Y)$ such that $\varphi(U) \cap X = U$. Then φ makes $X \leq Y$.

For $V \in \mathcal{T}^*(Y)$ let $\psi(V) = V \cap X$. Then ψ makes $Y \leq X$.

(c) Let f be a continuous function from Y onto X , and define $\varphi : \mathcal{T}^*(X) \rightarrow \mathcal{T}^*(Y)$ by $\varphi(U) = f^{-1}(U)$. Then φ makes $X \leq Y$.

(d) Let φ_i make $X_i \leq Y_i$ for $i \in I$, let $\mathcal{B}$ denote the canonical base for the cartesian product topology of X , and define $\varphi : \mathcal{B} \rightarrow \mathcal{T}^*(Y)$ by $\varphi(\bigcap_{i \in F} \pi_i^{-1}(U_i)) = \bigcap_{i \in F} \pi_i^{-1}(\varphi_i(U_i))$ for $F \in \mathcal{P}_\omega^*(I)$, $U_i \in \mathcal{T}^*(X_i)$ for $i \in F$. We have $X \leq Y$ from Lemma 2.15.

It is convenient to consider several classes of spaces.

Definition. Let X be a space, α an infinite cardinal, ρ an ordinal such that $\rho \leq \alpha$, and $2 \leq n < \omega$.

(a) X has the c.c.c. (or, X is a c.c.c. space) if every cellular family in X is finite or countably infinite, i.e., if $S(X) \leq \omega^+$.

(b) X is productively c.c.c. if $X \times Y$ is a c.c.c. space for every c.c.c. space Y .

(c) X has precalibre (α, ρ) if for every set $\{U_\xi : \xi < \alpha\}$ of non-empty, open subsets of X there is $A \subset \alpha$ such that $\text{type } A = \rho$ and $\{U_\xi : \xi \in A\}$

has the finite intersection property; X has *precalibre* α if X has precalibre (α, α) .

(d) X has *property* $K_{\alpha,n}$ if X has calibre (α, α, n) ; X has *property* K_n if X has property $K_{\omega^+, n}$; and X has *property* (K) if X has property K_2 .

It follows from Theorem 2.1(a) that a space with calibre ω^+ has property (K), and a space with property (K) has the c.c.c.

The notation c.c.c. abbreviates the expression *countable chain condition*.

2.17 Theorem. Let X and Y be spaces with $X \leq Y$. Then

- (a) $S(X) \leq S(Y)$;
- (b) if Y is productively c.c.c. then X is productively c.c.c.;
- (c) if α is an infinite cardinal and ρ an ordinal such that Y has precalibre (α, ρ) , then X has precalibre (α, ρ) ;
- (d) if α is an infinite cardinal, $2 \leq n < \omega$ and Y has property $K_{\alpha,n}$, then X has property $K_{\alpha,n}$.

Proof. Let $\varphi: \mathcal{T}^*(X) \rightarrow \mathcal{T}^*(Y)$ make $X \leq Y$.

(a) If the indexed family $\{U_i: i \in I\}$ is a cellular family in X , then $\{\varphi(U_i): i \in I\}$ is a cellular family in Y ; hence $S(X) \leq S(Y)$.

(b) Let Z be a c.c.c. space. Since $X \leq Y$ and $Z \leq Z$ we have $X \times Z \leq Y \times Z$ (from Theorem 2.16(c)) and hence

$$S(X \times Z) \leq S(Y \times Z) \leq \omega^+$$

from part (a). It follows that $X \times Z$ is a c.c.c. space, as required.

(c) For $\{U_\xi: \xi < \alpha\} \subset \mathcal{T}^*(X)$ we have $\{\varphi(U_\xi): \xi < \alpha\} \subset \mathcal{T}^*(Y)$ and since Y has precalibre (α, ρ) there is $A \subset \alpha$ with type $A = \rho$ such that $\{\varphi(U_\xi): \xi \in A\}$ has the finite intersection property. It follows that $\{U_\xi: \xi \in A\}$ has the finite intersection property, as required.

(d) For $\{U_\xi: \xi < \alpha\} \subset \mathcal{T}^*(X)$ we have $\{\varphi(U_\xi): \xi < \alpha\} \subset \mathcal{T}^*(Y)$ and since Y has property $K_{\alpha,n}$ there is $A \in [\alpha]^n$ such that if $C \subset A$ and $|C| = n$ then $\bigcap_{\xi \in C} \varphi(U_\xi) \neq \emptyset$. It follows that if $C \subset A$ and $|C| = n$ then $\bigcap_{\xi \in C} U_\xi \neq \emptyset$, as required.

Definition. For spaces X and Y , the notation $X \equiv Y$ means that $X \leq Y$ and $Y \leq X$.

It is clear that $\equiv$ is an equivalence relation on the class of spaces.

2.18 *Corollary.* Let X and Y be spaces with $X \equiv Y$. Then

- (a) $S(X) = S(Y)$;
- (b) X is productively c.c.c. if and only if Y is productively c.c.c.;
- (c) if α is an infinite cardinal and ρ an ordinal such that $\rho \leq \alpha$, then X has precalibre (α, ρ) if and only if Y has precalibre (α, ρ) ;
- (d) if α is an infinite cardinal and $2 \leq n < \omega$, then X has property $K_{\alpha, n}$ if and only if Y has property $K_{\alpha, n}$; and
- (e) $X \times Z \equiv Y \times Z$ for every space Z .

Proof. Parts (a), (b), (c) and (d) follow from the corresponding parts of Theorem 2.17, and (e) follows from Theorem 2.16(d).

The following simple criterion for $\equiv$ -equivalence, involving pseudobases, is useful.

2.19 *Lemma.* Let X and Y be spaces with pseudobases $\mathcal{B}$ and $\mathcal{C}$ respectively, and let φ be a function from $\mathcal{B}$ onto $\mathcal{C}$ such that if $n < \omega$ and $\{B_k : k < n\} \subset \mathcal{B}$, then $\bigcap_{k < n} B_k = \emptyset$ if and only if $\bigcap_{k < n} \varphi(B_k) = \emptyset$.

Then $X \equiv Y$.

Proof. Since $\varphi[\mathcal{B} \setminus \{\emptyset\}] \subset \mathcal{T}^*(Y)$ we have $X \leq Y$ from Lemma 2.15. For $V \in \mathcal{C}$ with $V \neq \emptyset$ we choose $\psi(V) \in \mathcal{B}$ such that $\varphi(\psi(V)) = V$; then $\psi(V) \neq \emptyset$ and again from Lemma 2.15 we have $Y \leq X$.

For a space X the Boolean algebra $\mathcal{R}(X)$ of regular-open subsets of X , and the Stone space $S(\mathcal{R}(X))$ of $\mathcal{R}(X)$ (which is the Gleason space $G(X)$ of X) are defined and discussed in the Appendix.

The next result shows that for every space X there is an extremally disconnected, compact Hausdorff space (namely, the Gleason space $G(X)$ of X) which is $\equiv$ -equivalent to X .

2.20 *Theorem.* If X is a space then $X \equiv G(X)$.

Proof. It follows from the Appendix B that the Boolean algebra $\mathcal{R}(X)$ is a pseudobase (indeed, a base) for $G(X)$. We define

$$\varphi : \mathcal{T}(X) \rightarrow \mathcal{R}(X) \subset \mathcal{T}(G(X))$$

by

$$\varphi(U) = \text{int}_X \text{cl}_X U \quad \text{for } U \in \mathcal{T}(X),$$

and we set $\varphi^* = \varphi|_{\mathcal{T}^*(X)}$. From Lemma B.14 of the Appendix it follows that

φ^* is a function from $\mathcal{T}^*(X)$ onto $\mathcal{R}^*(X)$, and if $n < \omega$ and $\{U_k : k < n\} \subset \mathcal{T}^*(X)$, then $\bigcap_{k < n} U_k = \emptyset$ if and only if $\bigcap_{k < n} \varphi(U_k) = \emptyset$.

Hence from Lemma 2.19 we have $X \equiv G(X)$, as required.

We continue with two consequences of Theorem 2.20.

2.21 Corollary. Let $\{X_i : i \in I\}$ be a set of spaces. Then $G(\prod_{i \in I} X_i) \equiv \prod_{i \in I} G(X_i)$.

Proof. From Theorems 2.20 and 2.16(d) we have

$$G\left(\prod_{i \in I} X_i\right) \equiv \prod_{i \in I} X_i \equiv \prod_{i \in I} G(X_i),$$

as required.

2.22 Corollary. Let α be an infinite cardinal, ρ an ordinal such that $\rho \leq \alpha$, $2 \leq n < \omega$, and X a space. Then

(a) X has precalibre (α, ρ) if and only if $G(X)$ has precalibre (α, ρ) ; and

(b) X has property $K_{\alpha, n}$ if and only if $G(X)$ has property $K_{\alpha, n}$.

Proof. The statements are immediate from Corollary 2.18 and Theorem 2.20.

The following simple proof derives directly from the definitions (and does not depend on the relations $\leq$ and $\equiv$ discussed above).

For $\lambda \geq \omega$ and X a space, the space $X_{(\lambda)}$ and its elementary properties are described in Appendix B.

2.23 Lemma. Let $\lambda > \omega$, let X be a locally compact Hausdorff space, and U a non-empty, open subset of $X_{(\lambda)}$. Then there are a non-empty, open subset V of $X_{(\lambda)}$, a cardinal $\mu \leq \lambda$ with $\mu < \lambda$ in case λ is a regular cardinal, and a set $\{V(\zeta) : \zeta < \mu\}$ of (non-empty) open subsets of X such that

$$V \subset U,$$

$$V = \bigcap_{\zeta < \mu} V(\zeta) = \bigcap_{\zeta < \mu} \text{cl}_X V(\zeta),$$

$$\text{cl}_X V(\zeta) \text{ is compact for } \zeta < \mu, \text{ and}$$

$$\{V(\zeta) : \zeta < \mu\} \text{ is closed under finite intersections.}$$

Proof. Let $p \in U$. It follows from Appendix B that there are $\mu \leq \lambda$, with $\mu \geq \omega$ and with $\mu < \lambda$ in case λ is regular, and a set $\{U(\zeta) : \zeta < \mu\}$ of open subsets of X , such that

$$p \in \bigcap_{\zeta < \mu} U(\zeta) \subset U;$$

we assume without loss of generality that $\text{cl}_X U(\zeta)$ is compact for $\zeta < \mu$.

We set $U(\zeta, 0) = U(\zeta)$, and if $n < \omega$ and $U(\zeta, n)$ has been defined we choose an open subset $U(\zeta, n+1)$ of X such that

$$p \in U(\zeta, n+1) \subset \text{cl}_X U(\zeta, n+1) \subset U(\zeta, n).$$

We note that $\text{cl}_X U(\zeta, n)$ is compact for all $n < \omega$. We let $\{V(\zeta) : \zeta < \mu\}$ be an indexing of the set of finite intersections of elements of the family

$$\{U(\zeta, n) : \zeta < \mu, n < \omega\},$$

and we set $V = \bigcap_{\zeta < \mu} V(\zeta)$. It is clear that V, μ and $\{V(\zeta) : \zeta < \mu\}$ are as required.

2.24 Corollary. Let $\alpha \geq \beta \geq \omega$ and $\lambda \geq \omega$, and let X be a locally compact Hausdorff space. If $X_{(\lambda)}$ has precalibre (α, β) , then $X_{(\lambda)}$ has calibre (α, β) .

Proof. Let $\{U_\xi : \xi < \alpha\}$ be a set of non-empty, open subsets of $X_{(\lambda)}$. We consider two cases.

Case 1. $\lambda = \omega$. For $\xi < \alpha$ let V_ξ be a non-empty, open subset of U_ξ such that $\text{cl}_X V_\xi$ is compact. There is $B \subset \alpha$ with $|B| = \beta$ such that $\{V_\xi : \xi \in B\}$ has the finite intersection property, and we have

$$\bigcap_{\xi \in B} U_\xi \supset \bigcap_{\xi \in B} \text{cl}_X V_\xi \neq \emptyset.$$

Case 2. $\lambda > \omega$. For $\xi < \alpha$ there are by Lemma 2.23 a non-empty open subset V_ξ of U_ξ , a cardinal $\mu(\xi) \leq \lambda$ and a set $\{V_\xi(\zeta) : \zeta < \mu(\xi)\}$ of open subsets of X such that

$$V_\xi = \bigcap_{\zeta < \mu(\xi)} V_\xi(\zeta) = \bigcap_{\zeta < \mu(\xi)} \text{cl}_X V_\xi(\zeta).$$

There is $B \subset \alpha$ with $|B| = \beta$ such that $\{V_\xi : \xi \in B\}$ has the finite intersection property, and since $\{\text{cl}_X V_\xi(\zeta) : \xi \in B, \zeta < \mu(\xi)\}$ has the finite

intersection property, we have

$$\bigcap_{\xi \in B} U_\xi \supset \bigcap_{\xi \in B} V_\xi = \bigcap \{ \text{cl}_X V_\xi(\zeta) : \xi \in B, \zeta < \mu(\xi) \} \neq \emptyset,$$

as required.

2.25 Corollary. Let $\alpha \geq \omega$ and let X be a space. If X has precalibre α , then X has precalibre $\text{cf}(\alpha)$.

Proof. From Corollary 2.18(b) and Theorem 2.20 it follows that $G(X)$ has precalibre α . Then $G(X)$ has calibre α (by Corollary 2.24), hence calibre $\text{cf}(\alpha)$ (by Theorem 2.3(a)), hence precalibre $\text{cf}(\alpha)$. Then from Corollary 2.18(b) and Theorem 2.20 it follows that X has precalibre $\text{cf}(\alpha)$, as required.

We note that Corollary 2.25 can be proved directly, without using Lemma 2.23 or the Gleason space $G(X)$, by an argument similar to that of Theorem 2.3(a).

Notes for Chapter 2

Calibres are the creation of the Soviet mathematician Shanin (Šanin); without his highly original papers (1946a, b, c, 1948) the present monograph could not have been written or even conceived. His work is described in Chapter 3 and its Notes; for the moment we note simply that the expression calibre α is first introduced (with very nearly the definition we have adopted here) in Shanin (1946b), and that the case $\alpha = \beta = \gamma$ of Theorem 2.5(a)(i) is in Shanin (1946b, Theorem 2).

Those properties which in our terminology are denoted calibre $(\omega^+, 2, 2)$ and calibre $(\omega^+, \omega^+, 2)$ were considered by Marczewski (1941) and Knaster (1945), respectively.

In considering chain conditions satisfied by a space X , many authors (see for example Juhász (1971, 1980) and Rudin (1975)) prefer to deal not with the Souslin number $S(X)$, but with the *cellular number* $c(X)$, defined by the rule:

$$c(X) = \sup \{ |\mathcal{U}| : \mathcal{U} \text{ is cellular family in } X \}.$$

In our experience such discussions usually require considering separately the case in which there is, and the case in which there is not, a cellular family $\mathcal{U}$ in X such that $|\mathcal{U}| = c(X)$ ('the sup = max

problem'); this explains our preference for the cardinal $S(X)$ over $c(X)$. In any event, the relation between the two is easily described. We have

$$S(X) = (c(X))^+ \text{ if there is cellular } \mathcal{U} \subset \mathcal{T}(X) \text{ with } |\mathcal{U}| = c(X)$$

$$S(X) = c(X) \text{ otherwise.}$$

Historically, the cellular number precedes what we have called the Souslin number. The cellular number was introduced and studied in the doctoral dissertation of Kurepa (1935).

So far as we are aware, the first mathematician to consider (what we have called) cellular families and a chain condition was Cantor (1882); he showed, without the concept of separability at his disposal, that the Euclidean spaces $\mathbb{R}^n$ are c.c.c. spaces. The famous problem of Souslin (1920) (discussed briefly in the Notes for Chapter 7) belongs also to the theory of chain conditions: 'Un ensemble ordonné linéairement sans sautes ni lacunes et tel que tout ensemble de ses intervalles (contenant plus qu'un élément) n'empiétant pas les uns sur les autres est au plus dénombrable, est-il nécessairement un continu linéaire (ordinaire)?'

Hewitt (1948) defined pseudocompact spaces (as those completely regular Hausdorff spaces on which every continuous real-valued function is bounded) and gave several characterizations based on the ring $C(X)$ and the inclusion $X \subset \beta X$. That Hewitt's spaces are exactly those for which every locally finite family of open subsets is finite was shown by Glicksberg (1952); see also Mardesić & Papić (1955) and Bagley, Connell & McKnight (1958). Isbell (1964) (page 135) introduced pseudo- α -compact spaces for $\alpha \geq \omega$ as those (uniform) spaces whose covering character in the fine uniformity is at most α ; it is noted by Noble (1969a) (page 389) that Isbell's pseudo- α -compact spaces are exactly ours. Pseudo- (α, β) -compact spaces were introduced (for $\alpha \geq \beta \geq \omega$) by Comfort & Negrepontis (1972b).

Frolík (1960) considered those spaces which (in our terminology) have compact-calibre ω , and he showed that the product of any such space with a pseudocompact space is pseudocompact. Noble (1969a) improves some of Frolík's results and gives analogues for countably compact spaces.

The properties calibre (α, β, γ) and compact-calibre (α, β, γ) were

defined in the abstract of Comfort & Negrepointis (1978a). See also Miščenko (1966a, b) for another (two-cardinal) generalization of Shanin's calibre (to a family of sets; in particular, to a topology on a set) and for applications to the question of expressing a uniformly continuous function $f: X_I \rightarrow Y$ on a product X_I in the form $f = g \circ \pi_J$ with $J \subset I, |J| \leq \omega$.

Comfort (1971, Theorem 4) and Comfort & Negrepointis (1974, Theorem 3.5) have recorded the following theorem of Haratomi (1931, Satz 21): If X is metrizable, then $S(X) = (d(X))^+$. The case $\lambda = 2$ of Theorem 2.14, a generalization of Haratomi's result due to Erdős & Tarski (1943), has been proved independently by several authors, in some cases with a view to the application of Boolean-valued models in set theory; see in this connection Vopěnka (1965), Štěpánek & Vopěnka (1967), Efimov (1968), Bukovský (1968, introduction to section 2), and, for a generalized treatment, Mostowski (1969, Chapter 13, Section 3). Our motivation for introducing the λ -Souslin numbers $S_\lambda(X)$ was the observation that the above-cited argument (the case $\lambda = 2$) sufficed with only minor modifications to prove Theorem 2.14 in its present degree of generality.

The material in the section 'Reduction to compact Hausdorff spaces', including in particular the definition of the relations $\leq$ and $\equiv$ and the useful result that $X \equiv G(X)$ (Theorem 2.20), is from Galvin (1980).